

HARSH NARAYAN SINGH

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Education

B.Tech in Computer Science Engineering (Data Science)

Ajay Kumar Garg Engineering College, Ghaziabad

2022 – 2026

CGPA: 7.71 / 10

Technical Skills

Languages	Python, C++, JavaScript, CSS
Web Technology	HTML, CSS, CSS Preprocessors, MVC, JSON
Core CS	Data Structures, Algorithms, OOP, DBMS, Operating Systems, Computer Networks
AI / ML	XGBoost, Scikit-learn, Computer Vision, MediaPipe, Feature Engineering
Backend	Django REST, FastAPI, Flask, REST APIs, JWT, RBAC, Caching
Databases	PostgreSQL, MySQL, MongoDB, Supabase
Cloud & DevOps	AWS (EC2), Azure App Service, Docker, GitHub Actions, CI/CD
Tools	Git, Linux, Postman, Pytest

Experience

Python Developer — Dogra Technologies Nov 2025 – Aug 2026

- Developed a computer-vision-based virtual clothing try-on system using MediaPipe, applying body landmark detection and pixel-based measurement estimation to improve garment-fit accuracy.
- Built an XGBoost-based dynamic pricing module that used demand and business signals to support data-driven price recommendations.
- Implemented authentication and access-control workflows, including Google Sign-In and role-based access control, and deployed services on AWS EC2 with Supabase/PostgreSQL.

Software Development Engineer Intern — V4K Entertainment Pvt. Ltd., New Delhi Jun 2025 – Aug 2025

- Built and delivered multiple Django REST APIs with JWT authentication and role-based access control.
- Optimized ORM queries and introduced caching, reducing P95 API latency by **20%**.
- Containerized backend services using Docker and set up CI pipelines (pytest, flake8) for automated testing.

Data Science Intern — CodSoft

Nov 2024 – Dec 2024

- Built and evaluated machine learning pipelines for prediction tasks, covering data preprocessing, feature engineering, and model selection.
- Deployed trained models via Flask-based REST APIs for inference.

Projects

Dynamic Pricing Engine (Multi-Restaurant POS Platform) — Python, XGBoost, FastAPI, PostgreSQL, AWS

- Built an ML-driven dynamic pricing engine using demand signals and external factors such as weather and local events to generate price recommendations.
- Integrated with Clover and Square POS systems to support configurable dynamic pricing across multiple restaurant accounts.
- Deployed on cloud infrastructure with CI/CD pipelines enabling automated testing and reliable rollouts.

Virtual Clothing Try-On System — Python, MediaPipe, Computer Vision, AWS

- Designed a real-time computer vision pipeline using MediaPipe for body landmark detection and pixel-to-real-world measurement estimation.
- Calculated body proportions from detected landmarks to improve clothing-fit accuracy; deployed the application on AWS for real-time inference.

HealthCloud: IoT-Enabled Cloud Platform for Wearable Health Monitoring — Spring Boot, React 19, Capacitor Android, Supabase PostgreSQL, BLE

- Developed a three-tier health monitoring platform with a Spring Boot backend, React dashboard, and Capacitor-based Android application.
- Integrated Bluetooth Low Energy (BLE) smartwatch communication and GPS-based location tracking, with an automated alert engine for abnormal health readings.
- Implemented JWT authentication, role-based access control, and multi-institution data isolation for secure multi-tenant usage.

Achievements & Certifications

- Solved 250+ algorithmic problems across LeetCode, HackerRank, and CodeChef (CodeChef Rating: 1741).
- Certifications: MongoDB Basics (Udemy), Python Programming (Infosys Springboard), Fundamentals of ML – Scikit-learn (Infosys Springboard)